

MANOUANE EST, QC, Canada

WMO: 715210

Lat: 50.656N

Lon: 70.532W

Elev: 497

StdP: 95.50

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.5	(c) -34.6	(d) -41.5	(e) 0.1	(f) -37.4	(g) -38.4	(h) 0.1	(i) -34.7	(j) 12.3	(k) -19.5	(l) 11.3	(m) -19.5	(n) 1.8	(o) 100	(p) 0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	25.7	17.4	23.9	16.6	22.1	15.8	19.4	22.9	18.2	21.6	17.2	20.4	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	14.0	20.7	17.1	12.9	19.6	16.0	12.0	18.6	57.5	22.6	53.7	21.8	50.3	20.4	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.9	8.6	-42.0	29.4	2.2	1.8	-43.6	30.7	-45.0	31.8	-46.2	32.8	-47.8	34.1
			-42.2	21.7	2.2	1.4	-43.8	22.7	-45.1	23.5	-46.4	24.3	-48.0	25.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.8	-19.8	-18.8	-11.3	-2.3	6.0	12.6	15.5	14.6	9.8	2.8	-5.1	-14.8		
	DBStd	13.71	8.26	7.99	7.90	5.56	4.52	4.03	2.78	2.94	3.97	3.99	5.52	7.69		
	HDD10.0	4416	923	807	659	369	139	17	0	2	49	226	454	767		
	HDD18.3	7005	1182	1041	918	619	382	178	97	122	256	482	704	1026		
	CDD10.0	472	0	0	0	0	16	95	169	144	45	2	0	0		
	CDD18.3	21	0	0	0	0	1	5	8	6	2	0	0	0		
	CDH23.3	213	0	0	0	0	15	64	81	43	10	0	0	0		
	CDH26.7	26	0	0	0	0	2	9	9	4	1	0	0	0		
	WSAvg	3.4	3.4	3.4	3.6	3.6	3.1	3.2	3.1	3.1	3.5	3.9	3.8	3.2		
	PrecAvg	922	48	40	47	60	70	91	124	117	108	82	77	57		
	PrecMax	1082	91	70	84	102	146	152	180	175	230	119	119	106		
(15) Precipitation	PrecMin	760	24	12	14	18	35	37	66	59	49	49	46	30		
	PrecStd	82	19	13	17	20	23	29	27	31	35	21	20	18		
	DB	2.6	2.8	8.7	14.2	25.3	28.0	27.9	27.1	24.7	17.2	9.1	2.1	2.1		
	MCWB	2.3	1.0	3.5	6.2	14.5	18.2	19.0	19.6	18.8	12.8	8.3	1.8	1.8		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-0.1	0.2	4.7	10.7	21.4	25.1	25.6	24.4	21.2	13.4	5.6	0.4	0.4		
	2% MCWB	-0.2	-0.7	1.3	4.6	12.8	16.5	17.3	17.9	16.8	11.0	4.7	0.2	0.2		
	5% DB	-4.1	-2.7	2.2	7.9	18.1	22.8	23.6	22.4	18.7	11.2	3.1	-1.4	-1.4		
	5% MCWB	-4.6	-3.7	0.2	3.3	10.7	15.1	16.7	16.9	15.3	9.3	2.3	-1.7	-1.7		
	10% DB	-7.1	-5.8	0.3	5.5	15.1	20.5	21.7	20.4	16.6	8.9	1.6	-3.6	-3.6		
	10% MCWB	-7.5	-6.6	-1.1	2.1	9.3	14.1	16.0	16.0	13.7	7.6	1.1	-4.0	-4.0		
	0.4% WB	1.6	1.2	4.6	7.4	16.8	20.3	21.0	20.7	19.9	13.9	8.8	2.0	2.0		
	0.4% MCDB	1.8	1.8	7.3	12.1	21.4	25.0	24.9	24.3	22.5	16.0	9.4	2.1	2.1		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-1.1	-1.0	2.1	5.3	13.7	18.3	19.1	19.2	17.8	11.8	5.0	0.2	0.2		
	2% MCDB	-0.9	0.1	3.8	9.1	18.9	22.3	22.7	22.5	20.3	13.1	5.3	0.3	0.3		
	5% WB	-5.2	-3.9	0.4	4.0	11.9	16.7	18.0	18.0	16.0	9.6	2.5	-1.8	-1.8		
	5% MCDB	-4.9	-2.9	1.9	7.3	17.0	20.5	21.7	20.8	18.3	10.7	2.8	-1.6	-1.6		
	10% WB	-8.0	-6.6	-1.3	2.6	10.1	15.2	17.0	16.8	14.2	7.5	1.1	-4.0	-4.0		
	10% MCDB	-7.6	-5.8	0.3	5.2	14.7	19.2	20.4	19.4	16.1	8.7	1.6	-3.7	-3.7		
(35) Mean Daily Temperature Range	MDBR	11.9	13.6	14.3	12.1	12.0	11.5	10.3	9.7	9.1	6.6	6.7	9.9	9.9		
	5% DB	11.4	11.2	14.2	15.3	18.5	15.5	14.2	13.4	13.1	11.1	7.3	8.0	8.0		
	5% MCWBR	11.4	10.3	11.2	10.1	10.3	7.8	7.2	7.4	8.5	8.4	6.4	8.1	8.1		
	5% WB	11.7	10.5	12.3	12.6	15.8	12.6	11.3	10.7	11.3	9.6	6.5	8.2	8.2		
	5% MCWBR	11.6	9.7	10.1	8.7	9.8	7.3	6.6	6.7	8.3	8.0	6.3	8.2	8.2		
(40) Clear-Sky Solar Irradiance	taub	0.241	0.252	0.281	0.322	0.340	0.368	0.380	0.359	0.342	0.310	0.278	0.240	0.240		
	taud	2.286	2.277	2.247	2.246	2.435	2.409	2.390	2.463	2.487	2.545	2.444	2.314	2.314		
	Ebn at Noon	809	896	926	919	916	889	870	875	854	825	761	754	754		
	Edh at Noon	65	86	109	123	108	112	112	99	87	68	57	54	54		
(44) All-Sky Solar Radiation	RadAvg	1.20	2.31	3.67	4.79	5.39	5.43	4.91	3.98	3.00	1.71	0.87	0.70	0.70		
	RadStd	0.14	0.20	0.33	0.36	0.39	0.44	0.43	0.24	0.23	0.17	0.09	0.10	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	+246	+262	N/A	N/A

Nomenclature: See separate page